

KG-MMRAG Retrieval Ablation

Hybrid vs Vector-only vs KG-only across MIRAGE

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Benchmarks: MMLU-Med, MedQA-US, MedMCQA, PubMedQA*, BioASQ-Y/N

Questions: 10/benchmark (total 50)

Generator: google/medgemma-4b-it (CoT reasoning ON)

Encoder: pubmedbert-base-embeddings

Corpus: MedRAG textbooks (3,000 chunks)

KG: Hetionet (47,031 nodes / 2,250,197 edges)

Best strategy by mean accuracy: kg (0.740)

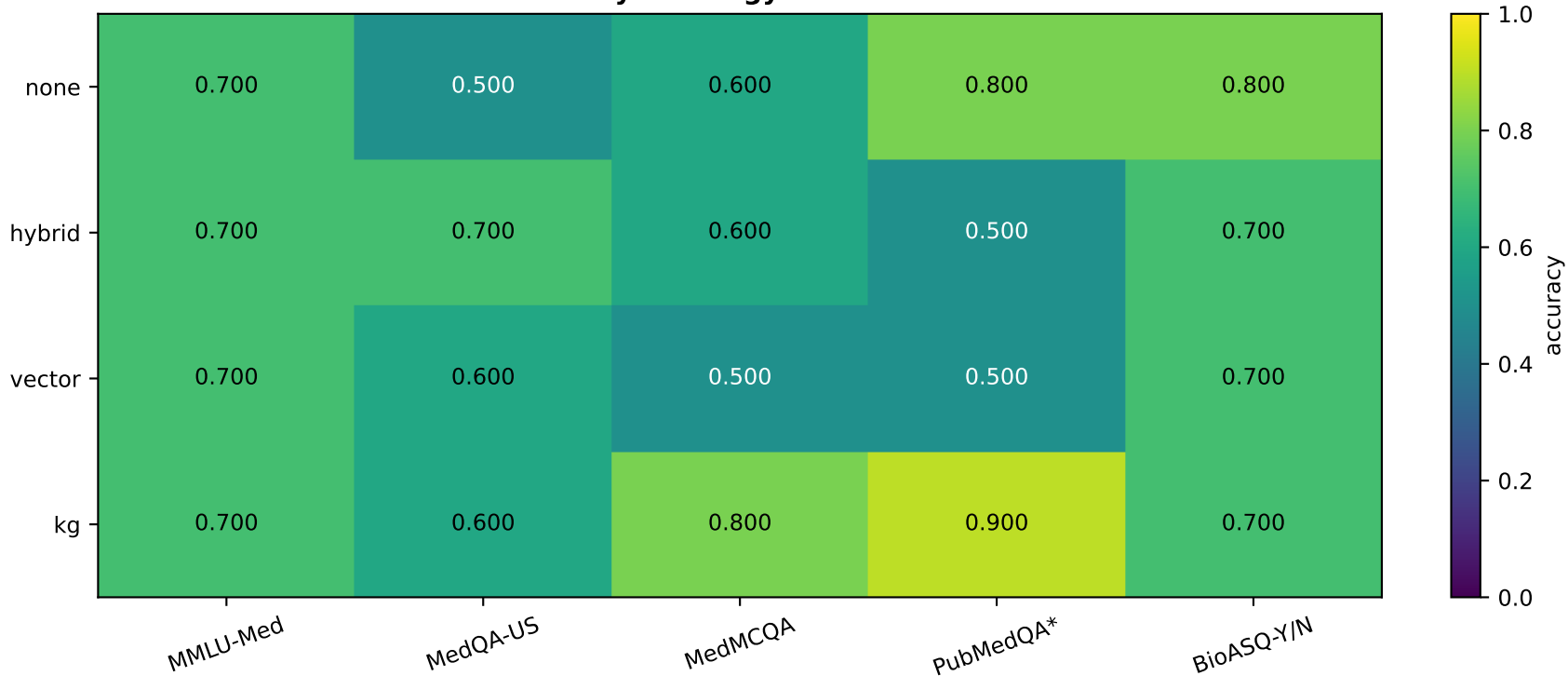
Retrieval gate: passages injected only if cosine ≥ 0.45 ;

KG facts relevance-ranked, kept only if cosine ≥ 0.35 (top 4).

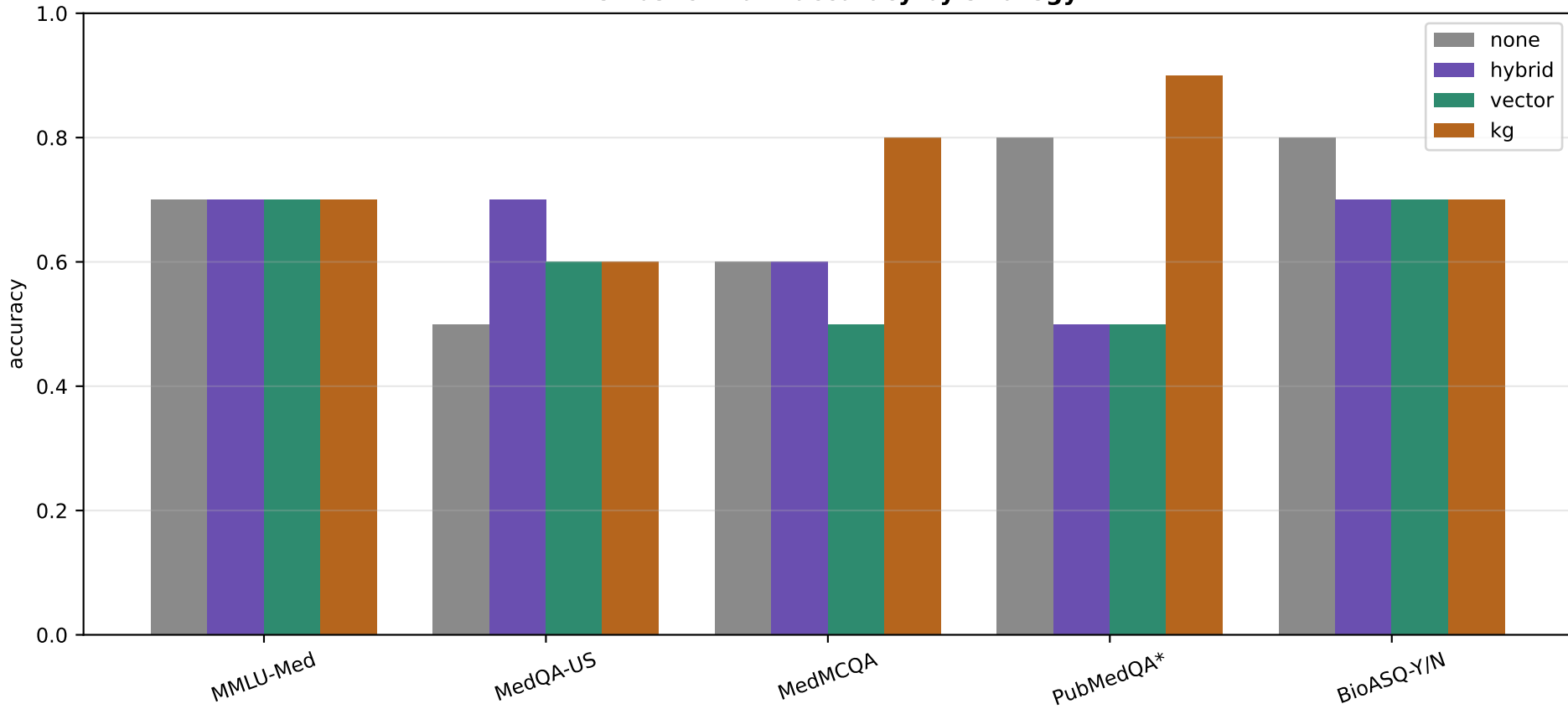
Gated-out retrieval falls back to closed-book for that question.

Note: retrieval quality is corpus-dependent (MIRAGE): benchmarks whose ideal corpus is absent here will show limited retrieval gains.

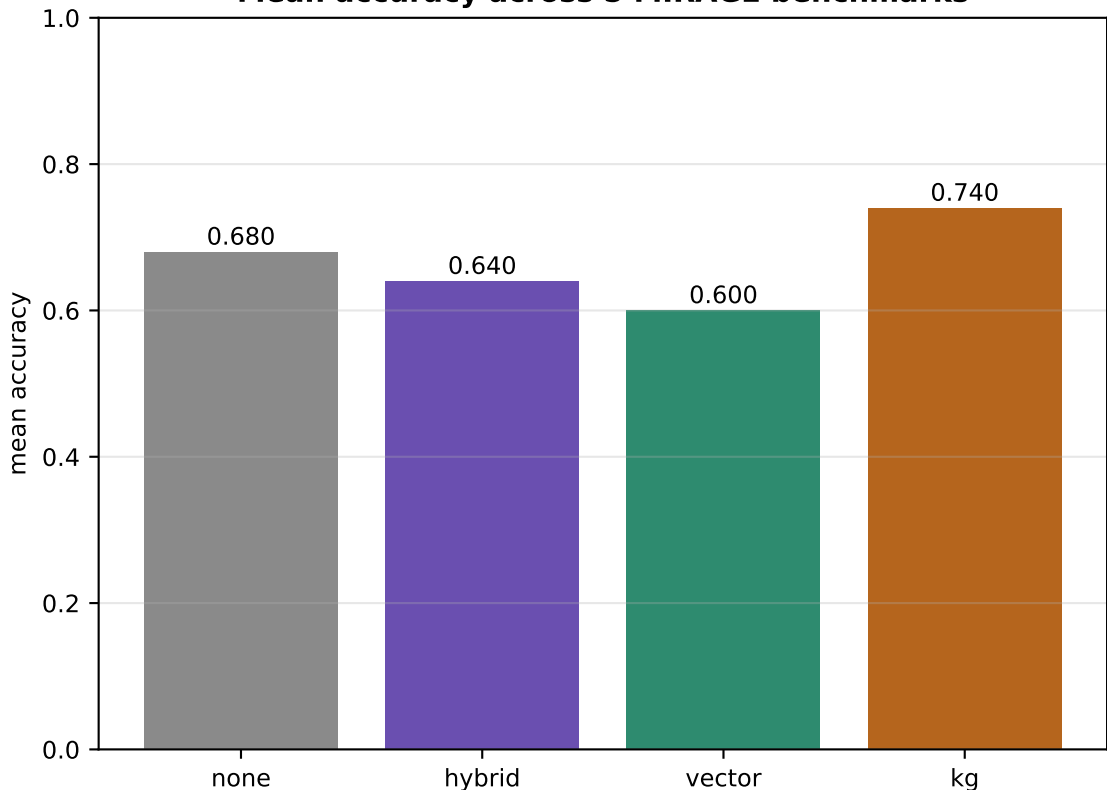
Accuracy: strategy x benchmark



Per-benchmark accuracy by strategy

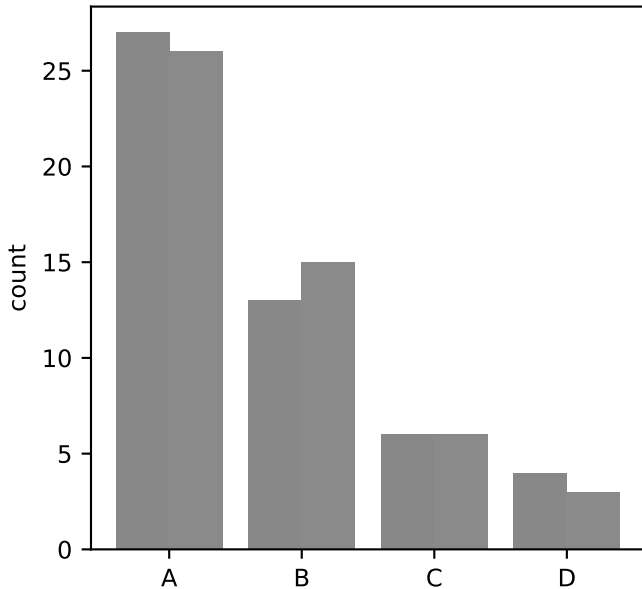


Mean accuracy across 5 MIRAGE benchmarks

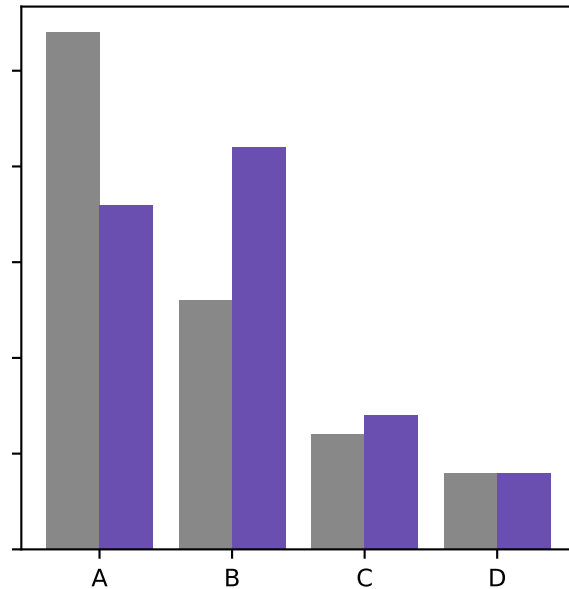


Answer-choice distribution (gold vs predicted) — checks letter bias

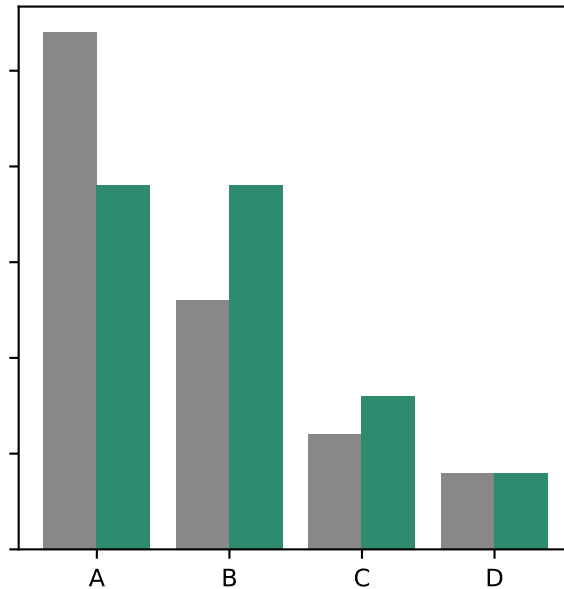
none



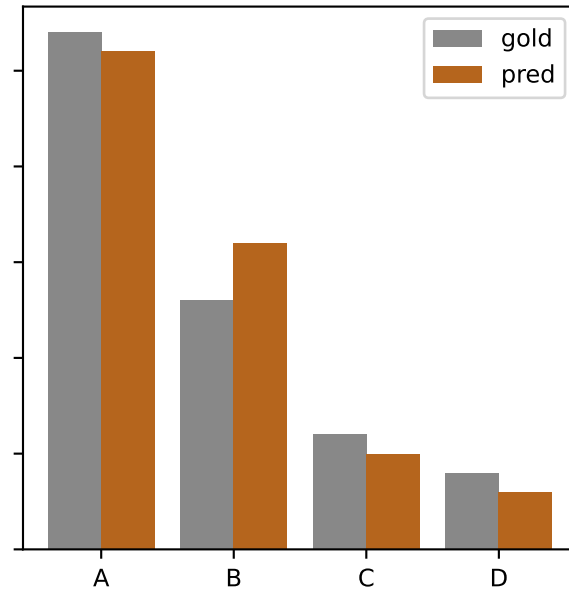
hybrid



vector



kg



Statistical comparison

McNemar paired test on per-question correctness (continuity-corrected, $df=1$). b_{01} = first wrong & second right; b_{10} = first right & second wrong.

$p < 0.05 \Rightarrow$ the two strategies differ significantly on that benchmark.

Key contrasts: each retrieval mode vs 'none' (does retrieval help at all?), and hybrid vs vector (does the KG add anything on top of passages?).

[MMLU-Med]

hybrid vs none $b_{01}= 0$ $b_{10}= 0$ $\chi^2= 0.00$ $p=1.0000$ (n.s.)

vector vs none $b_{01}= 0$ $b_{10}= 0$ $\chi^2= 0.00$ $p=1.0000$ (n.s.)

kg vs none $b_{01}= 0$ $b_{10}= 0$ $\chi^2= 0.00$ $p=1.0000$ (n.s.)

hybrid vs vector $b_{01}= 0$ $b_{10}= 0$ $\chi^2= 0.00$ $p=1.0000$ (n.s.)

[MedQA-US]

hybrid vs none $b_{01}= 0$ $b_{10}= 2$ $\chi^2= 0.50$ $p=0.4795$ (n.s.)

vector vs none $b_{01}= 0$ $b_{10}= 1$ $\chi^2= 0.00$ $p=1.0000$ (n.s.)

kg vs none $b_{01}= 1$ $b_{10}= 2$ $\chi^2= 0.00$ $p=1.0000$ (n.s.)

hybrid vs vector $b_{01}= 0$ $b_{10}= 1$ $\chi^2= 0.00$ $p=1.0000$ (n.s.)

[MedMCQA]

hybrid vs none $b_{01}= 2$ $b_{10}= 2$ $\chi^2= 0.25$ $p=0.6171$ (n.s.)

vector vs none $b_{01}= 2$ $b_{10}= 1$ $\chi^2= 0.00$ $p=1.0000$ (n.s.)

kg vs none $b_{01}= 0$ $b_{10}= 2$ $\chi^2= 0.50$ $p=0.4795$ (n.s.)

hybrid vs vector $b_{01}= 0$ $b_{10}= 1$ $\chi^2= 0.00$ $p=1.0000$ (n.s.)

[PubMedQA*]

hybrid vs none $b_{01}= 3$ $b_{10}= 0$ $\chi^2= 1.33$ $p=0.2482$ (n.s.)

vector vs none $b_{01}= 3$ $b_{10}= 0$ $\chi^2= 1.33$ $p=0.2482$ (n.s.)

kg vs none $b_{01}= 0$ $b_{10}= 1$ $\chi^2= 0.00$ $p=1.0000$ (n.s.)

hybrid vs vector $b_{01}= 0$ $b_{10}= 0$ $\chi^2= 0.00$ $p=1.0000$ (n.s.)

[BioASQ-Y/N]

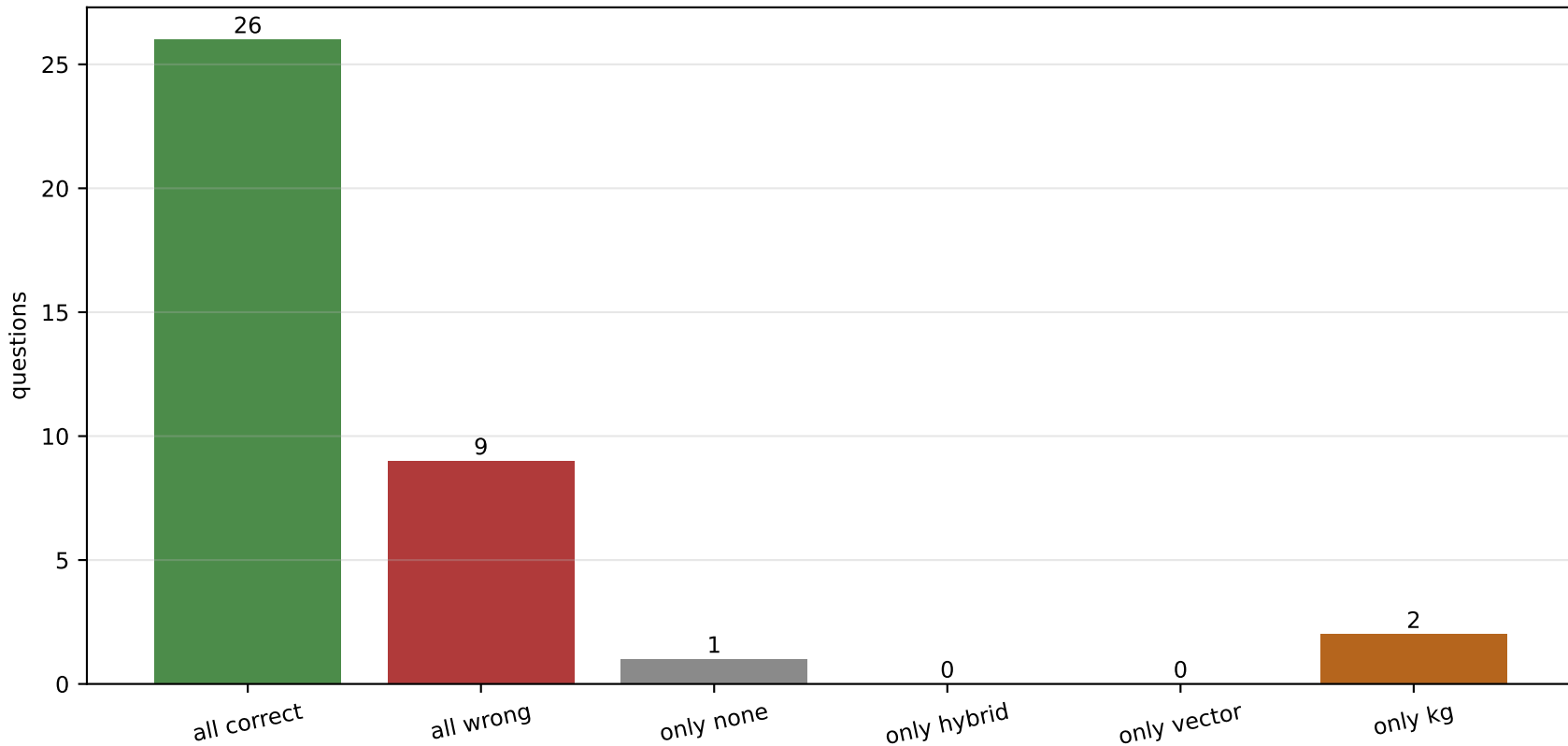
hybrid vs none $b_{01}= 2$ $b_{10}= 1$ $\chi^2= 0.00$ $p=1.0000$ (n.s.)

vector vs none $b_{01}= 1$ $b_{10}= 0$ $\chi^2= 0.00$ $p=1.0000$ (n.s.)

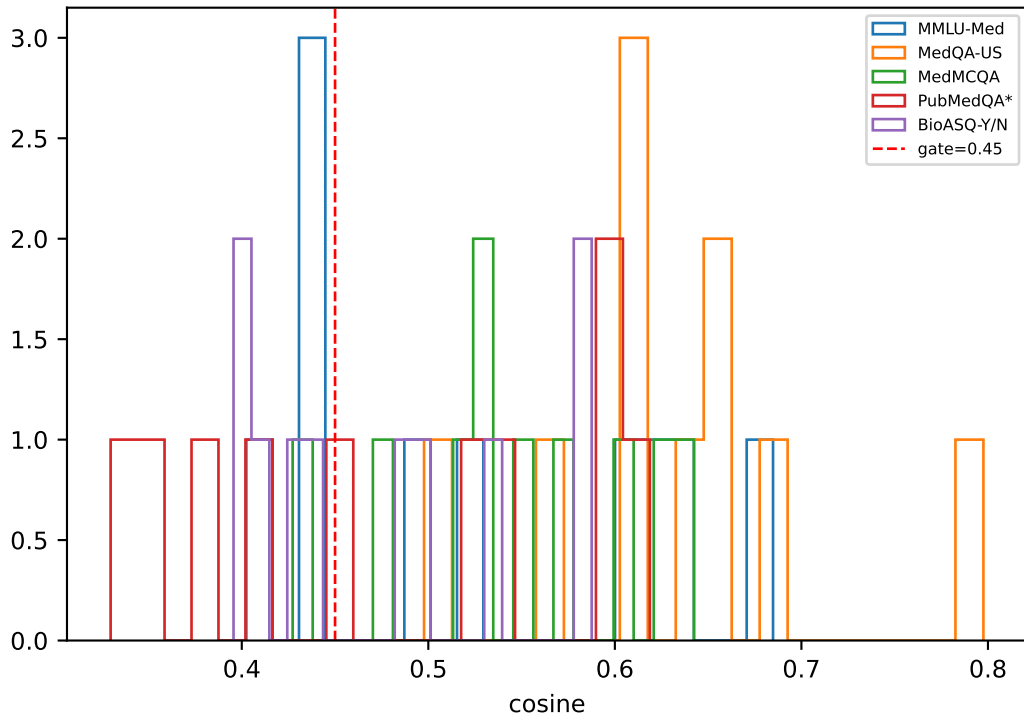
kg vs none $b_{01}= 2$ $b_{10}= 1$ $\chi^2= 0.00$ $p=1.0000$ (n.s.)

hybrid vs vector $b_{01}= 1$ $b_{10}= 1$ $\chi^2= 0.50$ $p=0.4795$ (n.s.)

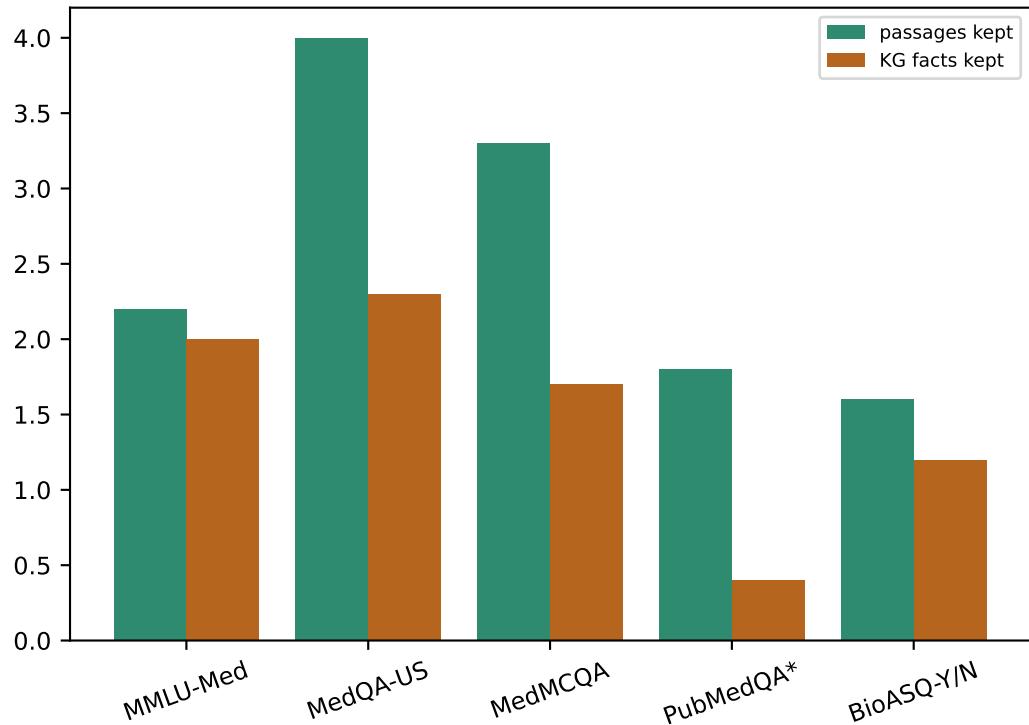
Agreement / disagreement across all questions



Vector: max top-k cosine (vs gate)



Context surviving the gate (avg per Q)



Error analysis — helped

Error analysis — KG FIXED a vector-only error

[MedQA-US] A 67-year-old man with transitional cell carcinoma of the bladder comes to the physician because of a 2-day history of r
gold=D hybrid=D vector=C kg=D

kg fact: brain cancer —presents→ Hearing Loss

[MedMCQA] Concentration of tropicamide:

gold=A hybrid=A vector=D kg=A

kg fact: Tropicamide —causes→ Keratitis

[BioASQ-Y/N] Can losartan reduce brain atrophy in Alzheimer's disease?

gold=B hybrid=B vector=A kg=B

kg fact: Galantamine —treats→ Alzheimer's disease

Error analysis — hurt

Error analysis — KG BROKE a correct vector answer

[BioASQ-Y/N] Does CIDEB mutation protect from liver disease?

gold=A hybrid=B vector=A kg=B

kg fact: RDH11 —participates→ Disease

Appendix — full metrics

Accuracy matrix (strategy x benchmark):

	MMLU-Med	MedQA-US	MedMCQA	PubMedQA*	BioASQ-Y/N	AVG
none	0.700	0.500	0.600	0.800	0.800	0.680
hybrid	0.700	0.700	0.600	0.500	0.700	0.640
vector	0.700	0.600	0.500	0.500	0.700	0.600
kg	0.700	0.600	0.800	0.900	0.700	0.740

Wall-clock seconds (benchmark x strategy):

MMLU-Med	none=70s	hybrid=98s	vector=81s	kg=68s
MedQA-US	none=115s	hybrid=123s	vector=138s	kg=125s
MedMCQA	none=113s	hybrid=136s	vector=142s	kg=103s
PubMedQA*	none=53s	hybrid=64s	vector=65s	kg=52s
BioASQ-Y/N	none=55s	hybrid=54s	vector=56s	kg=51s